

Infinitely many nontrivial root blocks in a Parseval operator-orbit frame

Candidate full solution packet for arXiv:2605.29671

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Status. Candidate full solution, likely valid, awaiting specialist review. For every fixed integer $m \geq 2$, we construct a Parseval single-operator orbit frame whose generator has infinitely many nontrivial root blocks of length m and a Riesz basis of generalized eigenvectors.

1 Source question

Gallardo-Gutiérrez and Partington study frames generated by operator orbits. In the model-space case, a pair (A, u) generating a frame is similar to (S_θ, k_0^θ) , where

$$K_\theta = H^2 \ominus \theta H^2, \quad S_\theta = P_{K_\theta} S|_{K_\theta}.$$

After discussing repeated zeros of the inner function θ , they write:

“It does not seem to be known whether generalized eigenvectors can form a Riesz sequence (if the zeros form a Carleson set and only finitely-many zeros are repeated, then this will happen), so we do not know whether it is possible to have infinitely-many nontrivial blocks if there is a frame.”

This appears on PDF page 17, Section 6.2(2), of arXiv:2605.29671 [1].

FRAME CONSTRUCTIONS ASSOCIATED WITH OPERATOR ORBITS

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- It does not seem to be known whether generalized eigenvectors can form a Riesz sequence (if the zeros form a Carleson set and only finitely-many zeros are repeated, then this will happen), so we do not know whether it is possible to have infinitely-many nontrivial blocks if there is a frame.
- (3) For the multiplication frame (M_φ, f) discussed in Section 3, we see that with φ an automorphism we are in situation (ii) above (similarity with the shift); the more complicated situation (i) does not occur.
- (4) As discussed in [14], on writing $\mathcal{H}_1$ for the Hardy operator on $L^2(0, 1)$ defined by

$$(\mathcal{H}_1 f)(x) = \frac{1}{x} \int_0^x f(t) dt, \quad 0 < x < 1$$

Figure 1: The open question on PDF page 17 of the source paper.

2 Proof intuition

An interpolating Blaschke product b has a Riesz basis of normalized kernels at its simple zeros. Raising b to a fixed power m repeats *every* zero m times, while the model space retains the

transparent orthogonal decomposition

$$K_{b^m} = K_b \oplus bK_b \oplus \cdots \oplus b^{m-1}K_b.$$

Thus $\{b^r e_j\}$ is a known Riesz basis. We compare it with the value and derivative kernels $\{d_{j,s} : 0 \leq s < m\}$, which are the root vectors of the backward shift.

The cross-Gram operator is upper triangular in the derivative order: a factor b^r has a zero of order r at every zero of b . Its diagonal blocks contain the normalized derivatives $(1 - |a_j|^2)b'(a_j)$, uniformly bounded away from zero by Carleson separation. Hence every diagonal block, and then the entire finite triangular operator, is invertible. This upgrades Besselness of the derivative kernels to a Riesz basis.

3 Preliminaries

We use the convention that the Hardy-space inner product is linear in its first variable. For $a \in \mathbb{D}$, let

$$D_{a,s}(z) = \frac{s!z^s}{(1 - \bar{a}z)^{s+1}}, \quad \langle f, D_{a,s} \rangle = f^{(s)}(a).$$

We need one standard sampling fact.

Lemma 1 (separated derivative sampling). *Let $\{a_j\} \subset \mathbb{D}$ be uniformly separated in the pseudo-hyperbolic metric. For every fixed $s \geq 0$, the family*

$$d_{j,s} := (1 - |a_j|^2)^{s+1/2} D_{a_j,s}$$

is Bessel in H^2 .

Proof. For $s = 0$, uniform separation makes $\sum_j (1 - |a_j|^2) \delta_{a_j}$ a Carleson measure, giving the assertion. For $s \geq 1$, choose pairwise disjoint pseudohyperbolic disks of one fixed small radius about the a_j . Subharmonicity on those disks gives

$$(1 - |a_j|^2)^{2s+1} |f^{(s)}(a_j)|^2 \leq C_s \int_{\Delta_j} |f^{(s)}(z)|^2 (1 - |z|^2)^{2s-1} dA(z).$$

Summing over the disjoint disks and using the standard weighted Hardy derivative estimate

$$\int_{\mathbb{D}} |f^{(s)}(z)|^2 (1 - |z|^2)^{2s-1} dA(z) \leq C'_s \|f\|_{H^2}^2$$

proves the Bessel inequality. This is the usual derivative form of Hardy sampling; the $s = 0$ interpolation statement is the Shapiro–Shields theorem [2]. $\square$

4 Construction and solution

Theorem 2. *For every fixed integer $m \geq 2$, there are an infinite-dimensional Hilbert space H , a contraction $A \in \mathcal{L}(H)$, and $u \in H$ such that:*

1. $\{A^n u : n \geq 0\}$ is a Parseval frame for H ;
2. A and A^* each have a Riesz basis of generalized eigenvectors;
3. there are infinitely many distinct eigenvalues, and the root block at each of them has length exactly m .

In particular, a frame can coexist with infinitely many nontrivial blocks.

Proof. For $j \geq 1$, put

$$a_j = \tanh(j+1).$$

The subtraction identity for the hyperbolic tangent gives

$$\rho(a_j, a_k) = \left| \frac{a_j - a_k}{1 - a_j a_k} \right| = \tanh |j - k|.$$

Moreover $\sum_j (1 - a_j) < \infty$, so these are the zeros of a Blaschke product b . If $p = \prod_{r=1}^{\infty} \tanh r$, then $p > 0$, since $\sum_r (1 - \tanh r) < \infty$. Therefore

$$\inf_j \prod_{k \neq j} \rho(a_j, a_k) \geq p^2 > 0.$$

Thus b is an interpolating Blaschke product. The normalized kernels

$$e_j(z) = \frac{(1 - a_j^2)^{1/2}}{1 - a_j z}$$

form a Riesz basis of K_b by the Shapiro–Shields theorem [2]; see also [3].

Set $\theta = b^m$, $H = K_\theta$, $A = S_\theta$, and $u = k_0^\theta$. The standard product decomposition of model spaces gives

$$K_{b^m} = \bigoplus_{r=0}^{m-1} b^r K_b.$$

Multiplication by each b^r is isometric, so

$$\mathcal{Y} = \{y_{j,r} := b^r e_j : j \geq 1, 0 \leq r < m\}$$

is a Riesz basis of K_θ .

For $0 \leq s < m$, define

$$d_{j,s} = (1 - a_j^2)^{s+1/2} D_{a_j, s}.$$

Every $d_{j,s}$ belongs to K_θ : if $h \in H^2$, then $(b^m h)^{(s)}(a_j) = 0$, hence $d_{j,s} \perp b^m H^2$. Lemma 1, applied finitely many times, shows that the combined family $\mathcal{D} = \{d_{j,s} : j \geq 1, 0 \leq s < m\}$ is Bessel.

Consider the cross-Gram operator from $\mathcal{D}$ to $\mathcal{Y}$, in blocks indexed by $0 \leq r, s < m$. Its entries are

$$\langle y_{i,r}, d_{j,s} \rangle = (1 - a_j^2)^{s+1/2} (b^r e_i)^{(s)}(a_j).$$

This vanishes when $s < r$, so the block matrix is upper triangular. On the diagonal $s = r$, Leibniz' rule gives

$$\begin{aligned} \langle y_{i,r}, d_{j,r} \rangle &= r! (1 - a_j^2)^{r+1/2} b'(a_j)^r e_i(a_j) \\ &= r! \beta_j^r \langle e_i, e_j \rangle, \quad \beta_j = (1 - a_j^2) b'(a_j). \end{aligned}$$

For a Blaschke product with simple zeros,

$$|\beta_j| = \prod_{k \neq j} \rho(a_j, a_k) \geq p^2,$$

while Schwarz–Pick gives $|\beta_j| \leq 1$. Consequently each diagonal block is a product of the invertible Gram operator of the Riesz basis $\{e_j\}$ and an invertible diagonal operator. The off-diagonal blocks are bounded because both $\mathcal{Y}$ and $\mathcal{D}$ are Bessel. A finite upper-triangular block operator with invertible diagonal blocks is invertible.

If $V_{\mathcal{Y}}$ and $V_{\mathcal{D}}$ are the two synthesis operators, the cross-Gram operator is $V_{\mathcal{Y}}^* V_{\mathcal{D}}$, up to the harmless transpose dictated by the inner-product convention. Since $V_{\mathcal{Y}}$ and this cross-Gram operator are invertible, $V_{\mathcal{D}}$ is invertible. Thus $\mathcal{D}$ is a Riesz basis of K_{θ} .

Directly from the displayed formula for $D_{a,s}$,

$$(S^* - \bar{a}I)D_{a,s} = sD_{a,s-1} \quad (s \geq 1), \quad (S^* - \bar{a}I)D_{a,0} = 0.$$

Since K_{θ} is S^* -invariant, $S_{\theta}^* = S^*|_{K_{\theta}}$. Hence $\mathcal{D}$ is a Riesz basis of root vectors for A^* , with a length- m block at every $\bar{a}_j = a_j$. In the normalized basis,

$$(A^* - a_j I)d_{j,s} = s(1 - a_j^2)d_{j,s-1},$$

and all these coefficients are nonzero. Writing $A^* = VLV^{-1}$ in this Riesz basis gives $A = V^{-*}L^*V^*$. The images under V^{-*} of the standard coordinate vectors are therefore a Riesz basis of root vectors for A , with the same block lengths.

It remains to check the frame. For $f \in K_{\theta}$, invariance under S^* and the reproducing property at zero give

$$\langle f, A^n u \rangle = \langle (A^*)^n f, k_0^{\theta} \rangle = (S^{*n} f)(0) = \widehat{f}(n).$$

Parseval's identity in H^2 now yields

$$\sum_{n \geq 0} |\langle f, A^n u \rangle|^2 = \sum_{n \geq 0} |\widehat{f}(n)|^2 = \|f\|_{H^2}^2.$$

Thus the orbit is a Parseval frame, completing the proof. $\square$

5 Scope, verification, and novelty

The theorem answers the literal existence question more strongly by allowing any prescribed uniform finite block length. There is a normalization caveat: the coefficients $s(1 - a_j^2)$ tend to zero. Thus the result does not claim a bounded similarity to the direct sum of canonical Jordan matrices having unit superdiagonal. It does prove infinitely many nontrivial root blocks and a Riesz basis of generalized eigenvectors, exactly the properties asked for.

The proof is analytic and uses no computer calculation. The principal review points are the standard derivative-sampling lemma and the orientation of the finite triangular cross-Gram operator. Both are recorded separately in `VERIFICATION.md`.

A bounded primary-source search through 12 August 2026 used the exact arXiv id and title together with the phrases *generalized eigenvectors*, *Riesz basis*, *model space*, *repeated zeros*, *derivative kernels*, and *infinitely many Jordan blocks*. It found background results on model-space derivatives but no exact resolution of the source question. Novelty confidence is moderate; specialist review is recommended before any priority claim.

References

- [1] E. A. Gallardo-Gutiérrez and J. R. Partington, *Frame constructions associated with operator orbits*, arXiv:2605.29671 (2026).
- [2] H. S. Shapiro and A. L. Shields, *On some interpolation problems for analytic functions*, Amer. J. Math. **83** (1961), 513–532.
- [3] S. R. Garcia, J. Mashreghi, and W. T. Ross, *Introduction to Model Spaces and Their Operators*, Cambridge Studies in Advanced Mathematics 148, Cambridge University Press, 2016.